

Efficient Eigen-Analysis for Large Delayed Cyber-Physical Power System Using Explicit Infinitesimal Generator Discretization

Hua Ye, *Member, IEEE*, Yutian Liu, *Senior Member, IEEE*, and Peng Zhang, *Senior Member, IEEE*

Abstract—Time delays significantly compromise the performance of wide-area measurement and control system and thus may jeopardize the stability of cyber-physical power systems (CPPS). A delayed CPPS (DCPPS) has a transcendental characteristic equation, leading to an infinite number of eigenvalues basically unsolvable by traditional eigen-analysis methods. In this paper, an explicit infinitesimal generator discretization (EIGD) approach is presented to tackle the traditionally intractable problem. First, the delayed differential equation of DCPPS is transformed to an ordinary differential equation by using an operator called infinitesimal generator. The operator is then optimally discretized, resulting in a highly structured, sparse and explicit approximant matrix. By exploiting the sparsity of the matrix and that of system matrices, the rightmost eigenvalues of the original DCPPS can be accurately computed. The contributions of the EIGD approach lie in the following: 1) it forms a theoretical foundation for accurately obtaining the critical eigenvalues of a CPPS with multiple delays; 2) it constructs a highly structured approximant matrix that enables efficient eigen-analysis of a large DCPPS by making full use of sparsity techniques; and 3) it integrates the shift-invert transformation, Arnoldi algorithm, Newton correction and eigen-sensitivity to form a computational framework for the analysis of large DCPPS. The accuracy, efficiency and scalability of EIGD have been extensively studied and thoroughly validated on the two-area four-machine test system and a practical large transmission grid.

Index Terms—Cyber-physical system, eigen-analysis, infinitesimal generator discretization, time delay, wide-area measurement system.

NOMENCLATURE

CPPS	Cyber-physical power system.
DCPPS	Delayed cyber-physical power system.

Manuscript received January 27, 2015; revised April 06, 2015 and June 24, 2015; accepted July 29, 2015. Date of publication August 14, 2015; date of current version April 15, 2016. This work was supported in part by the Natural Science Foundation of China under Grant 51107073, in part by the Specialized Research Fund for the Doctoral Program of Higher Education of China under Grant 11001311038, and in part by the U.S. National Science Foundation under Grant CNS-1419076. Paper no. TPWRS-00110-2015.

H. Ye is with the Key Laboratory of Power System Intelligent Dispatch and Control of Ministry of Education (Shandong University), Ji'nan 250061, China. He is also with the Department of Electrical and Computer Engineering, University of Connecticut, Storrs, CT 06269 USA (e-mail: yehua@sdu.edu.cn).

Y. Liu is with the Key Laboratory of Power System Intelligent Dispatch and Control of Ministry of Education (Shandong University), Ji'nan 250061, China (e-mail: liuyt@sdu.edu.cn).

P. Zhang is with the Department of Electrical and Computer Engineering, University of Connecticut, Storrs, CT 06269 USA (e-mail: peng@engr.uconn.edu).

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Digital Object Identifier 10.1109/TPWRS.2015.2463109

DDAE	Delayed differential and algebraic equation.
DDE	Delayed differential equation.
EIGD	Explicit infinitesimal generator discretization.
FACTS	Flexible ac transmission system.
HVDC	High voltage direct current.
IIGD	Implicit infinitesimal generator discretization.
LQR	Linear quadratic regulator.
ODE	Ordinary differential equation.
PMU	Phasor measurement unit.
PSS	Power system stabilizer.
WAMS	Wide-area measurement system.

I. INTRODUCTION

FOLLOWING the major blackouts in 1996 and 2003, WAMS using PMUs has been widely adopted in the North American power grids [1]. The U.S. Department of Energy's statistics indicate that 1126 PMUs have been installed as of 2013 [2] towards building future wide-area protection and control involving PSSs, HVDC systems and FACTS devices [3]–[5]. The power grids are therefore evolving into complex CPPS. A salient characteristic of CPPS, however, is the heterogeneous latency in transmitting and processing wide-area signals, leading to time delays ranging from tens to several hundred milliseconds [6]. According to a test in [7], the WAMS latency was found in a range between 250 and 740 ms. A recent study conducted by EPRI shows that the maximum latency of wide-area measurements may reach 460 ms [8]. Time delays can significantly jeopardize the performance of CPPS. Especially, for the wide-area damping control functions where control actions often need to be taken in the order of tens of milliseconds, time delays would weaken the damping effect causing sustained low frequency oscillations. Therefore, it is critical to accurately evaluate the impacts of time delays on power system small signal stability so that engineers can use the information in designing wide-area damping controllers adaptive to stringent time delays [9].

Lyapunov-Krasovskii functional technique has emerged as a time domain method to evaluate small signal stability with *single* communication delay. This approach, however, is inherently conservative and its accuracy is further compromised when model reduction is adopted for analyzing large CPPS [5]. On the other hand, frequency domain analysis (i.e.,

eigen-analysis), although traditionally dominating, faces significant challenges in considering time delays in DCPPS. This is because a DCPPS has a transcendental characteristic equation with an infinite number of eigenvalues basically unsolvable by traditional eigen-analysis methods. For DCPPS, it is therefore critical to reduce the infinite eigenvalue problem into a finite one. A typical approach is to recast the transcendental characteristic equation into an algebraic polynomial by substituting an exponential delay term with a first-order lead-lag block (known as Rekasius's substitution) [10]. This approach, however, can only accurately compute eigenvalues on the imaginary axis whereas it cannot catch eigenvalues elsewhere. In addition, Padé rational polynomial approximations to the exponential delay terms have been widely used in designing wide-area damping controllers considering time delay effects [11], [12]. However, the effectiveness and accuracy of Padé approximation have not been rigorously verified, making it prone to missing or spurious solutions.

This paper offers a different solution framework for effective eigen-analysis of large DCPPS based on an EIGD. The new method consists of two core techniques, viz., *infinitesimal generator reformulation* and *Chebyshev discretization*. First, the infinitesimal generator is used to transform the DDAEs (which can be further reduced to DDEs by eliminating algebraic variables) of DCPPS into an ODE on a Banach space [13]. Then, a set of discrete points distributed in the past time interval which is determined by time delays are selected as shifted and scaled zeros of a Chebyshev polynomial [14]. At these points, the infinite infinitesimal generator is optimally discretized, resulting in a *sparse, explicit and finite-dimensional* approximant matrix. Finally, sparse eigenvalue techniques, such as Arnoldi algorithm, combined with Newton correction, are implemented by exploiting sparsities in both the approximant matrix and power system matrices to accurately and efficiently compute critical eigenvalues of DCPPS.

The contributions of this paper are threefold. 1) EIGD lays the theoretical foundation for accurately computing the rightmost eigenvalues of a CPSS with multiple time delays. 2) A highly structured, sparse and explicit approximant matrix for the infinitesimal generator is built, enabling the use of sparsity techniques in eigen-analysis of a large DCPPS. 3) A computation framework integrating shift-invert transformation, Arnoldi algorithm, Newton's method and eigenvalue sensitivity has been established for analyzing complex DCPPS.

The rest of the paper is organized as follows. Section II formulates the eigenvalue problem for DCPPS. Section III establishes the theoretical foundation of EIGD. EIGD-based numerical algorithm is detailed in Section IV. The effectiveness of the presented approach is validated in Sections V and VI, followed by Section VII that concludes the paper.

II. SYSTEM MODELING AND PROBLEM FORMULATION

A. Modeling of DCPPS

Let $\mathbf{x}(t) \in \mathbb{R}^{n \times 1}$ be the state variable vector at current time t , consisting of state variables of dynamic components, e.g., generators and the associated excitation systems, HVDC systems and FACTS devices, as well as supplementary damping

controllers on them, etc. $\mathbf{y}(t) \in \mathbb{R}^{l \times 1}$ is the algebraic variable vector, i.e., bus voltages.

Intuitively, the dynamics in DCPPS should consist of two parts. The first part represents the connectivity and interaction of power system dynamic components and is free of time delay. The second part describes the interfaces between the components and wide-area damping controllers, which introduce time delays to the system. Remote measurements for observing inter-area oscillation modes, such as relative rotor angles and active power deviation of tie-lines, are normally selected as input signals for wide-area damping controllers. The controllers then generate control signals and feed them back to the physical power system. Those measurements and control signals can always be represented by $\mathbf{x}(t)$ and $\mathbf{y}(t)$. Therefore, mathematically, the dynamics of DCPPS around an equilibrium $(\mathbf{x}(0), \mathbf{y}(0))$ can be derived as follows:

$$\Delta \dot{\mathbf{x}}(t) = \mathbf{A}_0 \Delta \mathbf{x}(t) + \mathbf{B}_0 \Delta \mathbf{y}(t) + \sum_{i=1}^m (\mathbf{A}_i \Delta \mathbf{x}(t - \tau_i) + \mathbf{B}_i \Delta \mathbf{y}(t - \tau_i)) \quad (1)$$

where $\tau_i, i = 1, \dots, m$ are time delays. The maximum delay among τ_i is $\tau_{\max}$. $\mathbf{A}_i \in \mathbb{R}^{n \times n}$ and $\mathbf{B}_i \in \mathbb{R}^{n \times l}$ ($i = 0, \dots, m$) are linearized system matrices. Here, $\mathbf{A}_0$ is a block diagonal matrix, while $\mathbf{A}_i$ and $\mathbf{B}_i$ ($i \neq 0$) are very sparse.

With lumped-parameter model of transmission lines, power network can be represented as linear algebraic equations and are decoupled for different time instants [15]

$$\begin{cases} \mathbf{0} = \mathbf{C}_0 \Delta \mathbf{x}(t) + \mathbf{D}_0 \Delta \mathbf{y}(t) \\ \mathbf{0} = \mathbf{C}_i \Delta \mathbf{x}(t - \tau_i) + \mathbf{D}_i \Delta \mathbf{y}(t - \tau_i), i = 1, \dots, m \end{cases} \quad (2)$$

where $\mathbf{C}_i \in \mathbb{R}^{l \times n}$, $\mathbf{D}_i \in \mathbb{R}^{l \times l}$ ($i = 0, 1, \dots, m$) are linearized system matrices. In the steady state and for any instant t_0 , $\mathbf{x}(t_0) = \mathbf{x}(t_0 - \tau_i) = \mathbf{x}(0)$ and $\mathbf{y}(t_0) = \mathbf{y}(t_0 - \tau_i) = \mathbf{y}(0)$. Thus it can be concluded from (2) that $\mathbf{C}_i = \mathbf{C}_0$ and $\mathbf{D}_i = \mathbf{D}_0$, where $\mathbf{D}_0$ has the same structure as nodal admittance matrix except that each complex element $G_{ij} + jB_{ij}$ is expanded into a 2×2 real block $\begin{pmatrix} G_{ij} & -B_{ij} \\ B_{ij} & G_{ij} \end{pmatrix}$ [16].

Equations (1) and (2) form the model for small signal stability analysis of DCPPS.

B. Eigenvalue Problem of a DCPPS

Since $\mathbf{D}_i$ ($i = 0, \dots, m$) are non-singular, eliminating $\Delta \mathbf{y}(t)$ and $\Delta \mathbf{y}(t - \tau_i)$ in (1) using (2) yields the following DDEs:

$$\begin{cases} \Delta \dot{\mathbf{x}}(t) = \tilde{\mathbf{A}}_0 \Delta \mathbf{x}(t) + \sum_{i=1}^m \tilde{\mathbf{A}}_i \Delta \mathbf{x}(t - \tau_i), t > 0 \\ \Delta \mathbf{x}(t) = \Delta \mathbf{x}(0) \triangleq \boldsymbol{\varphi}, t \in [-\tau_{\max}, 0] \end{cases} \quad (3)$$

where $\tilde{\mathbf{A}}_0$ is a dense matrix of system states, and $\tilde{\mathbf{A}}_i$ ($i = 1, \dots, m$) are sparse matrices of delayed system states:

$$\tilde{\mathbf{A}}_0 = \mathbf{A}_0 - \mathbf{B}_0 \mathbf{D}_0^{-1} \mathbf{C}_0 \quad (4)$$

$$\tilde{\mathbf{A}}_i = \mathbf{A}_i - \mathbf{B}_i \mathbf{D}_0^{-1} \mathbf{C}_0. \quad (5)$$

The characteristic equation corresponding to (3) is

$$\left(\tilde{\mathbf{A}}_0 + \sum_{i=1}^m \tilde{\mathbf{A}}_i e^{-\lambda \tau_i} \right) \mathbf{v} = \lambda \mathbf{v} \quad (6)$$

where λ is an eigenvalue and $\mathbf{v} \in \mathbb{R}^{n \times 1}$ is the corresponding eigenvector.

Equation (6) can be rewritten in an augmented form

$$\begin{bmatrix} \mathbf{A}'(\lambda) & \mathbf{B}'(\lambda) \\ \mathbf{C}_0 & \mathbf{D}_0 \end{bmatrix} \begin{bmatrix} \mathbf{v} \\ \mathbf{w} \end{bmatrix} = \mathbf{0} \quad (7)$$

where $\mathbf{w} \in \mathbb{R}^{l \times 1}$ is an ancillary vector and

$$\mathbf{A}'(\lambda) = \mathbf{A}_0 - \lambda \mathbf{I}_n + \sum_{i=1}^m \mathbf{A}_i e^{-\lambda \tau_i} \quad (8)$$

$$\mathbf{B}'(\lambda) = \mathbf{B}_0 + \sum_{i=1}^m \mathbf{B}_i e^{-\lambda \tau_i}. \quad (9)$$

The small signal stability of DCPPS can be analyzed by using Lyapunov's first method [17]. If all eigenvalues of (6) and (7) have negative real parts, the DCPPS (3) is asymptotically or small signal stable at $(\mathbf{x}(0), \mathbf{y}(0))$. If there exists an eigenvalue with a positive real part, the system is unstable.

However, both (6) and (7) are transcendental and have an infinite number of eigenvalues, which are basically unsolvable by traditional eigen-analysis methods. To resolve the challenge, an EIGD approach for computing critical eigenvalues of large DCPPS is presented.

III. THEORETICAL FOUNDATION: EIGD-BASED EIGEN-ANALYSIS

The fundamental idea of EIGD is to transform the DDEs of DCPPS into an ODE, and then obtain a set of critical eigenvalues from an approximant matrix corresponding to the ODE's coefficient. It is detailed as follows.

A. Problem Reformulation Using Infinitesimal Generator

Since the state variables $\mathbf{x}(t)$ are continuously differentiable for any $t \in [-\tau_{\max}, \infty)$, it can be proved that (3) can be reformulated as an abstract ODE on a Banach space [13]:

$$\begin{cases} \dot{\mathbf{u}}(t) = \mathcal{A} \mathbf{u}(t), t > 0 \\ \mathbf{u}(0) = \varphi \end{cases} \quad (10)$$

where the coefficient $\mathcal{A}$ is called an *infinitesimal generator* of a strongly continuous semigroup and $\mathbf{u}(t)$ is a continuously differentiable function. The solution of (10) relates to that of (3) by $\mathbf{u}(t) = \Delta \mathbf{x}(t + \theta)$, $\theta \in [-\tau_{\max}, 0]$.

According to the Spectral Mapping Principle [18], eigenvalues of $\mathcal{A}$ equal to those of the original DCPPS, which means one can compute eigenvalues of the DCPPS from $\mathcal{A}$ instead. However, finding eigenvalues of $\mathcal{A}$ is an infinite-dimensional problem on a Banach space. Therefore, one needs to discretize $\mathcal{A}$ first, and then obtain a reduced set of eigenvalues from a finite-dimensional approximant matrix.

B. Chebyshev Discretization of Infinitesimal Generator

To discretize $\mathcal{A}$, one can first form a mesh Ω_N with $N + 1$ distinct discretization points on the interval $[-\tau_{\max}, 0]$, where

N nonzero grid points are chosen as the scaled and shifted zeros of an N th-order Chebyshev polynomial of the second kind $U_N(\cdot)$ [14]. Equation (10) can then be discretized over Ω_N , which transforms the eigenvalue problem of (3) into a generalized eigenvalue problem. Solving the generalized eigenvalue problem gives an approximant matrix $\mathcal{A}_N$ of $\mathcal{A}$ [14]:

$$\mathcal{A}_N = \Pi_N^{-1} \Sigma_N \quad (11)$$

where Π_N and $\Sigma_N \in \mathbb{R}^{(N+1)n \times (N+1)n}$ are a companion-type matrix and a block upper-triangular matrix, respectively

$$\Pi_N = \frac{\tau_{\max}}{4} \begin{bmatrix} \frac{4}{\tau_{\max}} & \frac{4}{\tau_{\max}} & \frac{4}{\tau_{\max}} & \cdots & \frac{4}{\tau_{\max}} \\ 2 & 0 & -1 & & \\ & \frac{1}{2} & 0 & & \\ & & \frac{1}{3} & \ddots & \\ & & & \ddots & \frac{-1}{N-2} \\ & & & & 0 & \frac{-1}{N-1} \\ & & & & & \frac{1}{N} & 0 \end{bmatrix} \otimes \mathbf{I}_n$$

$$\Sigma_N = \begin{bmatrix} \mathbf{R}_0 & \mathbf{R}_1 & \cdots & \mathbf{R}_N \\ & \mathbf{I}_n & & \\ & & \ddots & \\ & & & \mathbf{I}_n \end{bmatrix}.$$

Here $\otimes$ denotes Kronecker product and $\mathbf{R}_j$ ($j = 0, \dots, N$) can be explicitly represented with $\tilde{\mathbf{A}}_i$ ($i = 0, \dots, m$)

$$\mathbf{R}_j = \tilde{\mathbf{A}}_0 + \sum_{i=1}^m \tilde{\mathbf{A}}_i T_j \left(-2 \frac{\tau_i}{\tau_{\max}} + 1 \right). \quad (12)$$

Notice that $T_j(\cdot)$ is the j th-order Chebyshev polynomial of the first kind.

Now the infinite-dimensional $\mathcal{A}$ is reduced into a finite-dimensional $\mathcal{A}_N$. This process is called the *EIGD*, through which one can solve a reduced set of critical eigenvalues of the DCPPS from $\mathcal{A}_N$ without having to solve the infinite eigenvalue problem.

C. Features of EIGD

EIGD has the following unique features:

- *It is explicit.* When analyzing a large DCPPS, $(\mathcal{A}_N)^{-1}$, i.e., $(\Sigma_N)^{-1} \Pi_N$, should be solved for eigenvalues. Fortunately, $(\Sigma_N)^{-1}$ has the same block structure as Σ_N . The i th element of the first block row of $(\Sigma_N)^{-1}$ can be explicitly expressed as $(\mathbf{R}_0)^{-1} \mathbf{R}_i$, $i = 0, \dots, N$. This explicitly allows one to fully exploit the sparsity of system matrices and makes it particularly suitable for solving large DCPPS, as can be seen from (4), (5), and (12). Actually, this is how the name *EIGD* got its name.
- *Σ_N and Π_N are highly structured.* The number of non-zero entries in Π_N is $(3N - 1)n$ and that of Σ_N is less than $(N + 1)n^2 + Nn$, while the total number of elements for each matrix is $(N + 1)^2 n^2$. Since $N \ll n$, both Π_N and Σ_N are very sparse.

These features guarantee the high efficiency of EIGD in processing large DCPPS. On the contrary, traditional IGD approach [19]–[21] featured by an implicit approximant matrix (denoted as *IIGD*) suffers from huge computational burden

and high memory requirement and is only applicable to small DCPSS.

Besides, the accuracy of EIGD is theoretically guaranteed [22]; in particular, the calculated eigenvalue of $\mathcal{A}_N$ has a low error bound of $\mathcal{O}(N^{-N})$.

Another important property is that all those inaccurate and diverged eigenvalues calculated from $\mathcal{A}_N$ always locate to the left of the accurate (converged) ones. This means that EIGD can always capture the most important eigenvalues locating on the right side.

In summary, the new approach has guaranteed efficiency, accuracy and scalability for large DCPSS.

IV. SCALABLE EIGD FOR ANALYZING LARGE DCPSS

Enabling the scalability of EIGD is of critical importance because in the real world, many DCPSS are large complex systems. The calculation of eigenvalues for already large DCPSS is even more challenging. To achieve the scalability, four major techniques are introduced.

A. Shift-Invert Transformation

Normally, sparse eigenvalue techniques aim at finding a group of eigenvalues with largest moduli [23]. However, critical eigenvalues that determine the small signal stability of DCPSS are always close to the imaginary axis leading to much smaller moduli than those of other eigenvalues. To facilitate the use of sparse eigenvalue techniques, the shift-invert transformation is applied to the approximant matrix $\mathcal{A}_N$ to shift the desirable eigenvalues so that they become dominant in moduli.

Suppose one would like to compute a set of eigenvalues close to a number s . First, all eigenvalues are shifted by $-s$. This can be achieved by replacing λ with $\lambda' + s$ in (6):

$$\left(\tilde{\mathbf{A}}'_0 + \sum_{i=1}^m \tilde{\mathbf{A}}'_i e^{-\lambda' \tau_i} \right) \mathbf{v} = \lambda' \mathbf{v} \quad (13)$$

where

$$\begin{aligned} \lambda' &= \lambda - s \\ \tilde{\mathbf{A}}'_0 &= \mathbf{A}_0 - s\mathbf{I}_n - \mathbf{B}_0 \mathbf{D}_0^{-1} \mathbf{C}_0 \\ \tilde{\mathbf{A}}'_i &= \mathbf{A}_i e^{-s\tau_i} - \mathbf{B}_i e^{-s\tau_i} \mathbf{D}_0^{-1} \mathbf{C}_0. \end{aligned}$$

After the shift operation, the approximant matrix $\mathcal{A}_N$ becomes $\mathcal{A}'_N$. The inverse of $\mathcal{A}'_N$ can be written as

$$(\mathcal{A}'_N)^{-1} = (\Sigma'_N)^{-1} \Pi_N \quad (14)$$

where the inverse of Σ'_N can be explicitly expressed by

$$\begin{aligned} (\Sigma'_N)^{-1} &= \begin{bmatrix} (\mathbf{R}'_0)^{-1} & -(\mathbf{R}'_0)^{-1} \mathbf{R}'_1 & \dots & -(\mathbf{R}'_0)^{-1} \mathbf{R}'_N \\ & \mathbf{I}_n & & \\ & & \ddots & \\ & & & \mathbf{I}_n \end{bmatrix} \\ \mathbf{R}'_0 &= \tilde{\mathbf{A}}'_0 + \sum_{i=1}^m \tilde{\mathbf{A}}'_i = \mathbf{A}'(s) - \mathbf{B}'(s) \mathbf{D}_0^{-1} \mathbf{C}_0 \\ \mathbf{R}'_j &= \tilde{\mathbf{A}}'_0 + \sum_{i=1}^m \tilde{\mathbf{A}}'_i T_j \left(-2 \frac{\tau_i}{\tau_{\max}} + 1 \right), j = 1, \dots, N. \end{aligned}$$

Notice that $\mathbf{A}'(s)$ and $\mathbf{B}'(s)$ can be obtained by replacing λ with s in (8) and (9), respectively.

B. Sparse Eigenvalue Computation Using Arnoldi Algorithm

The implicitly restarted Arnoldi algorithm [24] is employed to compute a specified number of eigenvalues with largest moduli from $(\mathcal{A}'_N)^{-1}$. In the algorithm, the most expensive operation is the matrix-vector product to generate Krylov sequences. Let $\mathbf{q}_k \in \mathbb{R}^{(N+1)n \times 1}$ be the k th Krylov sequence, then $\mathbf{q}_{k+1}$ can be computed as listed in Algorithm 1.

Algorithm 1 Matrix-vector product $\mathbf{q}_{k+1} = (\mathcal{A}'_N)^{-1} \mathbf{q}_k$

- 1: Compute $\mathbf{r} = \Pi_N \mathbf{q}_k$ by sparse matrix multiplication
 - 2: Reshape $\mathbf{r}$ to a matrix $\mathbf{V} = [\mathbf{v}_0, \dots, \mathbf{v}_N]$ in the column-wise direction, where $\mathbf{v}_j \in \mathbb{R}^{n \times 1}, j = 0, 1, \dots, N$
 - 3: $\mathbf{z} \leftarrow \mathbf{v}_0$
 - 4: **for** $j = 1, \dots, N$ **do**
 - 5: $\mathbf{p} \leftarrow \tilde{\mathbf{A}}'_0 \mathbf{v}_j$
 - 6: $\mathbf{w} \leftarrow \sum_{i=1}^m \tilde{\mathbf{A}}'_i T_j (-2(\tau_i/\tau_{\max}) + 1) \mathbf{v}_j$
 - 7: $\mathbf{z} \leftarrow \mathbf{z} - \mathbf{p} - \mathbf{w}$
 - 8: **end for**
 - 9: Compute $\mathbf{q}_{k+1}(1:n) = (\mathbf{R}'_0)^{-1} \mathbf{z}$
 - 10: $\mathbf{q}_{k+1}((n+1):(N+1)n) \leftarrow \mathbf{r}((n+1):(N+1)n)$
 - 11: **return** $\mathbf{q}_{k+1}$
-

In Algorithm 1, steps 5 and 9 are the most computationally intensive. To cut down the burden, the power method and inverse power method [25] are applied to the two steps (see the Appendix), respectively, by fully exploiting the sparsity of system matrices. In total, the computational burden of Algorithm 1 is nearly $N + 1$ times of the burden of computing the same number of eigenvalues of the system without delays.

Once an eigenvalue λ'' is obtained from $(\mathcal{A}'_N)^{-1}$, the eigenvalue of $\mathcal{A}_N$ can be transformed back by $\lambda = (1/\lambda'') + s = \lambda' + s$. The eigenvector $\mathbf{v}$ corresponding to λ can be directly estimated by the first n entries of the Krylov sequence [14].

C. Newton Correction

The eigenvalues and eigenvectors obtained in Section IV-B are still estimates of true solutions. Therefore, Newton's method is applied to search for true solutions.

Equation (7) can be linearized into

$$\begin{aligned} & \begin{bmatrix} \mathbf{A}'(\lambda^{(k)}) & \mathbf{B}'(\lambda^{(k)}) \\ \mathbf{C}_0 & \mathbf{D}_0 \end{bmatrix} \begin{bmatrix} \Delta \mathbf{v}^{(k)} \\ \Delta \mathbf{w}^{(k)} \end{bmatrix} \\ & - \begin{bmatrix} \mathbf{v}^{(k)} + \sum_{i=1}^m \tau_i e^{-\lambda^{(k)} \tau_i} (\mathbf{A}_i \mathbf{v}^{(k)} + \mathbf{B}_i \mathbf{w}^{(k)}) \\ \mathbf{0} \end{bmatrix} \Delta \lambda^{(k)} \\ & = \mathbf{J}'^{(k)} \begin{bmatrix} \Delta \mathbf{v}^{(k)} \\ \Delta \mathbf{w}^{(k)} \end{bmatrix} + \mathbf{g}'^{(k)} \Delta \lambda^{(k)} = -\mathbf{f}^{(k)} \end{aligned} \quad (15)$$

where $\lambda^{(k)}, \mathbf{v}^{(k)}$ and $\mathbf{w}^{(k)}$ are estimates of eigenvalue, eigenvector and ancillary vector at the k th iteration, respectively, and

$\mathbf{f}^{(k)} \in \mathbb{R}^{n \times 1}$ denotes the deviation of the augmented characteristic equation at the k th iteration.

The correction vector can then be obtained

$$\begin{bmatrix} \Delta \mathbf{v}'^{(k)} \\ \Delta \mathbf{w}^{(k)} \end{bmatrix} = -(\mathbf{J}^{(k)})^{-1} \mathbf{f}^{(k)} + \frac{(\mathbf{J}^{(k)})^{-1} \mathbf{g}^{(k)} \mathbf{e}_1^T (\mathbf{J}^{(k)})^{-1} \mathbf{f}^{(k)}}{1 + \mathbf{e}_1^T (\mathbf{J}^{(k)})^{-1} \mathbf{g}^{(k)}}. \quad (16)$$

Here $\Delta \mathbf{v}'^{(k)}$ is identical to $\Delta \mathbf{v}^{(k)}$ except that the first entry is replaced by $\Delta \lambda^{(k)}$. Similarly, $\mathbf{J}^{(k)}$ is identical to $\mathbf{J}'^{(k)}$ except that the first column is replaced by $\mathbf{e}_1$, the first column of an identity matrix of order $n + l$. Therefore, $\mathbf{J}^{(k)}$ is as sparse as $\mathbf{J}'^{(k)}$. $\mathbf{g}^{(k)}$ equals to $\mathbf{g}'^{(k)}$ minus $\mathbf{e}_1$.

The stop criteria of Newton correction is

$$\max \left(\|\Delta \mathbf{v}'^{(k)}\|, \|\Delta \mathbf{w}^{(k)}\| \right) < \varepsilon_1 \text{ or } \|\mathbf{f}^{(k)}\| < \varepsilon_1$$

where ε_1 is a small tolerance.

Actually, the number of Newton iterations can be used as a good metric for evaluating the accuracy of EIGD.

D. Eigenvalue Sensitivity to Time Delay

Some eigenvalues of DCPPS are more sensitive to the variation of time delays, while others are less sensitive because they are more correlated to system states at the current time. A new metric is introduced to distinguish these two kinds of eigenvalues as follows:

$$\frac{\partial \lambda}{\partial \tau_i} = \mathbf{u}^T \frac{\partial \left(\tilde{\mathbf{A}}_0 + \sum_{i=1}^m \tilde{\mathbf{A}}_i e^{-\lambda \tau_i} \right)}{\partial \tau_i} \mathbf{v} = -\lambda e^{-\lambda \tau_i} \mathbf{u}^T \tilde{\mathbf{A}}_i \mathbf{v} \quad (17)$$

where $\mathbf{u} \in \mathbb{R}^{n \times 1}$ is the left eigenvector associated with λ .

The eigenvalue sensitivity metric can also be used to quantify the impact of time delays in the cyber network on the small signal stability of DCPPS.

E. Flowchart of EIGD

A computational flowchart is given in Fig. 1 to show the general procedure for analyzing large DCPPS with EIGD. It consists of five main steps, i.e., specifying parameters, modeling the DCPPS, Arnoldi iteration, Newton correction, and eigenvalue sensitivity analysis.

V. CASE STUDY I: TWO-AREA FOUR-MACHINE SYSTEM

The well-known two-area four-machine test system [26] as depicted in Fig. 2 is extensively studied to validate the correctness, accuracy and efficiency of EIGD. In all test cases, two constant wide-area time delays are considered.

A. Modeling of the System

The system is operating with area 1 exporting 400 MW to area 2. Each generator is equipped with a thyristor exciter with high transient gain and a PSS using rotor speed as input. A wide-area damping controller is installed on G_1 in area 1. The controller has the same lead-lag structure as a PSS and uses the relative rotor speed between G_1 and G_3 (in area 2) as the wide-area feedback signal. The gain of the controller $K_s = 20$ and the time constant of the wash-out block $T_w = 5$ s. The time constants of two lead-lag blocks are $T_1 = T_3 = 0.0824$ s and

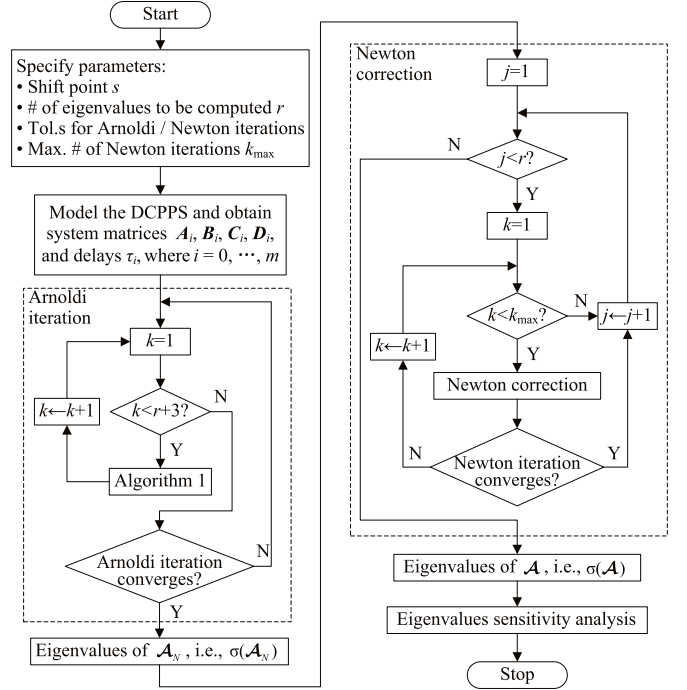


Fig. 1. Flowchart of EIGD.

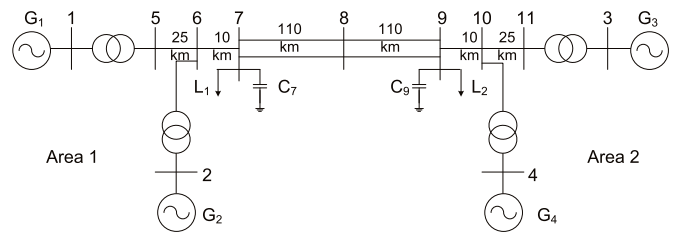


Fig. 2. One-line diagram of the two-area four-machine test system.

$T_2 = T_4 = 0.0303$ s, respectively. Time delays in feedback and output of the controller are assumed to be $\tau_1 = 120$ ms and $\tau_2 = 100$ ms, respectively. There are 56 state variables and 22 algebraic variables. Other data for the system can be found in [26].

All tests are carried out using MATLAB on an Intel 3.4-GHz computer with 8 GB of RAM. The number of guard-vectors for the implicitly restarted Arnoldi algorithm is $r + 3$, where r is the number of eigenvalues to be computed. The tolerance for the algorithm is 10^{-6} . In Newton correction, the maximum allowed number of iterations is 20 and the tolerance $\varepsilon_1 = 10^{-6}$.

B. Correctness of $\mathcal{A}_N$ in Approximating $\mathcal{A}$

Since constructing the approximant matrix $\mathcal{A}_N$ is the foundation of EIGD, it is critical to validate the correctness of $\mathcal{A}_N$ in approximating $\mathcal{A}$. This is achieved by comparing critical eigenvalues of $\mathcal{A}_N$ against those of $\mathcal{A}$.

For a large N of 50, the size of $\mathcal{A}_N$ is $(N + 1) \cdot n = 2856$. All eigenvalues of $\mathcal{A}_N$, i.e., $\sigma(\mathcal{A}_N)$, are obtained through QR algorithm. Then, those eigenvalues are used as initial guesses for the Newton's method so that a reduced set of accurate eigenvalues of $\mathcal{A}$, i.e., $\sigma(\mathcal{A})$, are obtained (see Fig. 3 and column 2 of Table I).

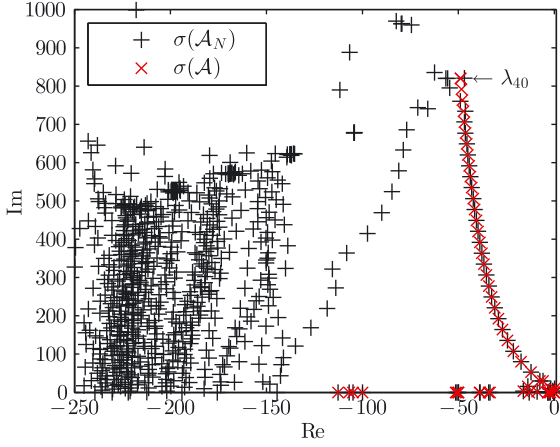


Fig. 3. Eigenvalues of $\mathcal{A}_N$ in case $N = 50$. They converge to those of $\mathcal{A}$ in the region of $\text{Re} \in [-47, 0]$ and $\text{Im} \in [0, 705]$ as well as the neighborhood of $(-100, 0)$. Other eigenvalues of $\mathcal{A}_N$ are diverged and locate to the left of the converged ones.

TABLE I
EIGENVALUES OF $\mathcal{A}$ AND THEIR SENSITIVITIES TO TIME DELAYS

i	λ_i	$\partial \lambda_i / \partial \tau_1$	Iter. no.*	
1	-2.088708	1.2394	0	0
2	-3.275351	-0.0406	1	0
3	-3.398661	-0.1843	0	0
4	-13.062606	4.7522	0	1
5	-8.947273 ± j 1.812823	-21.6064 ± j 24.1017	0	1
6	-16.143071 ± j 3.355718	0.0034 ± j 0.0063	0	1
7	-0.788883 ± j 4.020935	0.3726 ± j 1.4736	0	0
8	-1.936159 ± j 8.640199	0.0306 ± j 0.0526	0	0
9	0.080502 ± j 9.832372	11.1516 ± j 26.0961	0	1
10	-13.866119 ± j 15.523111	-2.7967 ± j 4.4067	1	1
11	-5.869500 ± j 30.102904	-395.9981 ± j 252.5447	1	1
12	-7.582637 ± j 29.881004	33.3728 ± j 38.8610	1	2
13	-12.181679 ± j 53.061871	1001.4599 ± j 36.5168	1	1
14	-17.385826 ± j 79.847387	2209.6236 ± j 283.9323	1	1
15	-21.328077 ± j 107.595623	3953.1657 ± j 623.9680	2	2
16	-24.437198 ± j 135.726204	6239.0235 ± j 1020.9911	2	2
17	-26.987700 ± j 164.031559	9067.7418 ± j 1457.7970	2	2
18	-29.144623 ± j 192.427175	12440.2002 ± j 1922.7996	2	2
19	-31.011163 ± j 220.873759	16357.1786 ± j 2406.5957	2	2
20	-32.655296 ± j 249.351057	20819.0844 ± j 2901.1035	2	-
21	-34.123912 ± j 277.847783	25825.9794 ± j 3399.2196	2	-
22	-35.450625 ± j 306.357269	31377.6677 ± j 3894.5953	2	-
23	-36.660313 ± j 334.875382	37473.7689 ± j 4381.4676	2	-
24	-37.771879 ± j 363.399464	44113.7667 ± j 4854.5316	3	-
25	-38.800016 ± j 391.927753	51297.0396 ± j 5308.8451	1	-
26	-39.756360 ± j 420.459046	59022.8798 ± j 5739.7567	3	-
27	-40.650284 ± j 448.992508	67290.5043 ± j 6142.8522	3	-
28	-41.489444 ± j 477.525747	76099.0628 ± j 6513.9147	3	-
29	-42.280176 ± j 506.063736	85447.6416 ± j 6848.8933	3	-
30	-43.027777 ± j 534.600764	95335.2673 ± j 7143.8796	4	-
31	-43.736725 ± j 563.138400	105760.9085 ± j 7395.0893	3	-
32	-44.410838 ± j 591.676473	116723.4779 ± j 7598.8479	3	-
33	-45.053394 ± j 620.214855	128221.8331 ± j 7751.5791	3	-
34	-45.667232 ± j 648.753448	140254.7779 ± j 7849.7964	3	-
35	-46.254827 ± j 677.292180	152821.0633 ± j 7890.0950	9	-
36	-46.818346 ± j 705.830993	165919.3884 ± j 7869.1469	4	-
37	-47.359700 ± j 734.369845	179548.4014 ± j 7783.6956	6	-
38	-47.880582 ± j 762.908704	193706.7002 ± j 7630.5527	6	-
39	-48.382500 ± j 791.447545	208392.8336 ± j 7406.5950	8	-
40	-48.866800 ± j 819.986349	223605.3022 ± j 7108.7615	6	-

* The first and second sub-columns contain the required number of Newton iterations for eigenvalues of $\mathcal{A}_N$ computed by QR and EIGD for $N = 50$, respectively.

It can be seen from Fig. 3 that in the region of $\text{Re} \in [-47, 0]$ and $\text{Im} \in [0, 705]$ as well as the neighborhood of $(-100, 0)$, the

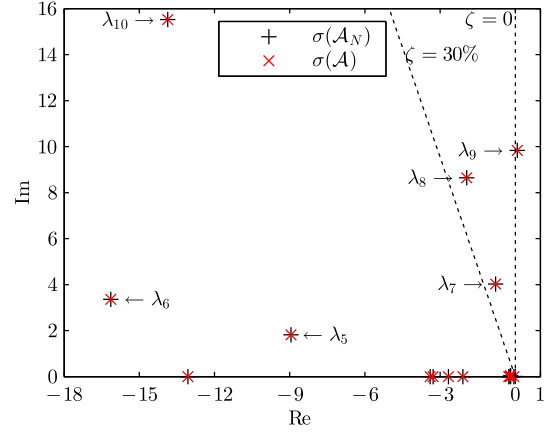


Fig. 4. Zoomed-in view of Fig. 3. Two dotted lines are composed of points with constant damping ratios $\zeta = 0$ and 30% , respectively. All electro-mechanical oscillation modes in the range of 0.1 to 2.5 Hz, i.e., $\lambda_7 - \lambda_9$, are accurately computed by EIGD. They are sufficient for small signal stability analysis of DCPSS.

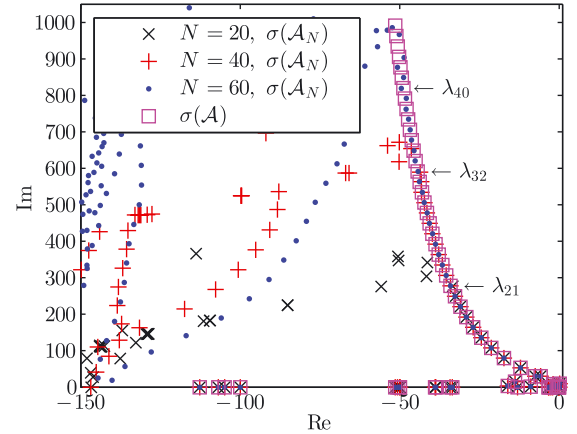


Fig. 5. Eigenvalues computed by EIGD from $\mathcal{A}_N$ in cases $N = 20, 40$ and 60 . When $N = 20$ and 40 , eigenvalues up to λ_{21} and λ_{32} can be accurately computed. When $N = 60$, six more eigenvalues can be captured than those with $N = 50$.

eigenvalues of $\mathcal{A}_N$ coincide with those of $\mathcal{A}$. Other eigenvalues of $\mathcal{A}_N$ are diverged and locate to the left of the converged ones. Fortunately, those rightmost eigenvalues of $\mathcal{A}_N$ are sufficient already for small signal stability analysis of DCPSS, as seen from Fig. 4.

The accuracy of a converged eigenvalue can be measured by the number of Newton iterations to obtain it. As can be seen in column 4 of Table I, all eigenvalues converge within 10 iterations. Among the converged eigenvalues, those with higher imaginary parts ($\lambda_{11} - \lambda_{40}$) need more iterations. In fact, those eigenvalues are significantly influenced by delays (see eigenvalue sensitivities to time delays in column 3 of Table I), leading to lower accuracy. Moreover, $\lambda_{11} - \lambda_{40}$ have high frequencies beyond the range of electro-mechanical oscillation modes and do not affect small signal stability of DCPSS, which is a desirable feature of EIGD.

The final step of this test is to examine the relationship between the accuracy of $\mathcal{A}_N$ and the granularity of Chebyshev discretization. As seen from Fig. 5, when $N = 20$ and 40 , eigenvalues up to λ_{21} and λ_{32} can be accurately captured already.

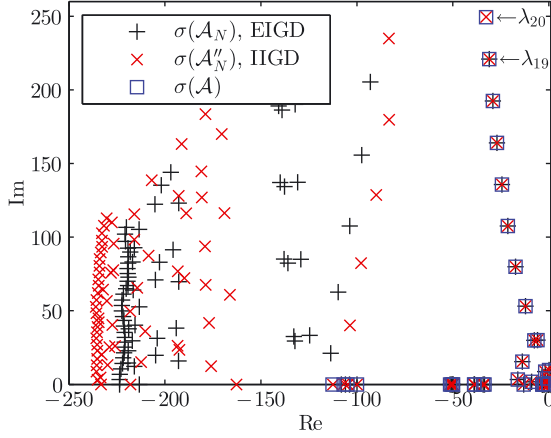


Fig. 6. Eigenvalues computed by EIGD and IIGD from $\mathcal{A}_N$ and $\mathcal{A}_N''$ in case $N = 50$. The rightmost and accurate ones are nearly identical, except that IIGD can obtain one extra eigenvalue λ_{20} of $\mathcal{A}$ than EIGD.

TABLE II
COMPARISON OF THE THREE METHODS
[# OF ARNOLDI ITERATIONS/CPU TIME (S)]

Methods	$N=20$	$N=40$	$N=50$	$N=60$
EIGD	2 / 6.24	2 / 10.61	2 / 12.69	2 / 15.21
IIGD	2 / 10.14	2 / 25.84	2 / 34.96	2 / 45.81
QR	- / 1.97	- / 13.71	- / 27.06	- / 47.70

When using a very fine granularity of $N = 60$, only six more eigenvalues can be captured than those with $N = 50$. Obviously, a low granularity of $N = 20$ results in an $\mathcal{A}_N$ sufficiently approximating $\mathcal{A}$, while further increasing N does not provide any more useful information.

C. Accuracy of EIGD

Now, we illustrate the accuracy of EIGD. Given $N = 50$ and a small shift point $s = 0.05$, EIGD can enumerate $r = 200$ eigenvalues of $\mathcal{A}_N$ close to s , as shown in Fig. 6. After less than 2 Newton iterations (see column 5 of Table I), EIGD obtains the most critical eigenvalues $\lambda_1 - \lambda_{19}$. Those rightmost eigenvalues are sufficient for the analysis of DCPPS.

The accuracy of EIGD is further compared with that of IIGD [19]–[21]. Assume the approximant matrix of $\mathcal{A}$ for IIGD is $\mathcal{A}_N''$. As shown in Fig. 6, $\sigma(\mathcal{A}_N)$ and $\sigma(\mathcal{A}_N'')$ computed by EIGD and IIGD respectively are nearly identical, except that IIGD can obtain one extra eigenvalue λ_{20} of $\mathcal{A}$ than EIGD.

D. Efficiency of EIGD

Table II summarizes computational time of EIGD, IIGD and QR. As seen in the table, the computational complexity of EIGD and N have a linear relationship. This is mainly because the computational burden of matrix-vector product involved in the Arnoldi algorithm is normally proportional to $N + 1$. However, the computational burden of both IIGD and QR rises exponentially as N increases. For high performance computation with $N = 40, 50$ and 60 , EIGD becomes more efficient over both IIGD and QR.

Notice that for small power system with very low granularity (e.g., $N = 20$), QR may be more efficient than EIGD even it

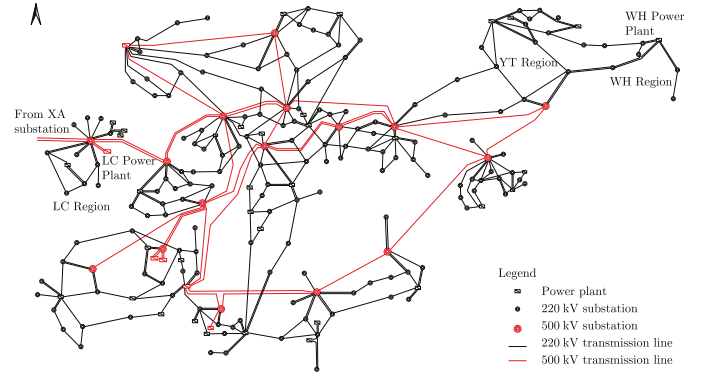


Fig. 7. Backbone transmission system.

computes all eigenvalues of $\mathcal{A}_N$. This is because QR is effective and robust for small sized dense matrices. When the system becomes large or high accuracy analysis is needed, however, EIGD will be found much more efficient than IIGD and QR.

VI. CASE STUDY II: A REAL-LIFE LARGE TRANSMISSION SYSTEM

A real-life large transmission system is undertaken for further test of EIGD. The backbone of the system is shown in Fig. 7. Modeling of the system is described in Section VI-A. In Section VI-B, the scalability of EIGD is verified where two constant wide-area time delays are considered. In Section VI-C, non-deterministic time delays are simulated and the impact of uncertainties in delays on small signal stability is analyzed. Sensitivities of the most critical eigenvalues to time delays are studied in Section VI-D.

A. Modeling of the System

The system has 516 buses, 114 generators, 936 transformers and transmission lines, and 299 loads. It has a weakly damped inter-area mode, where generators in the LC region oscillate against those in the YT and WH regions. Two identical wide-area LQRs [4] are installed on units #1 and #2 in LC power plant. Feedback signals for the two controllers are transmitted from unit #3 in WH plant to units #1 (channel 1) and to unit #2 (channel 2) in the LC plant, consisting of relative rotor speed and relative rotor angle. The gains for the relative rotor speed and relative rotor angle are 40 and 0.125, respectively. The total delay of channel 1 is $\tau_1 = 90$ ms and that of channel 2 is $\tau_2 = 100$ ms. Overall, there are 1128 state variables and 4637 algebraic variables.

Parameters for Arnoldi algorithm and Newton correction are the same as those described in Section V-A.

B. Scalability of EIGD

Here N is chosen as 25 and the size of $\mathcal{A}_N$ becomes $(N + 1) \cdot n = 29328$. Two shift points $s = j7$ and $j13$ are selected because we aim at finding electromechanical oscillation modes. For each shift point, we specify r to be 50 and 100 for comparison purposes. Fig. 8 shows the computed eigenvalues and Table III summarizes computational times.

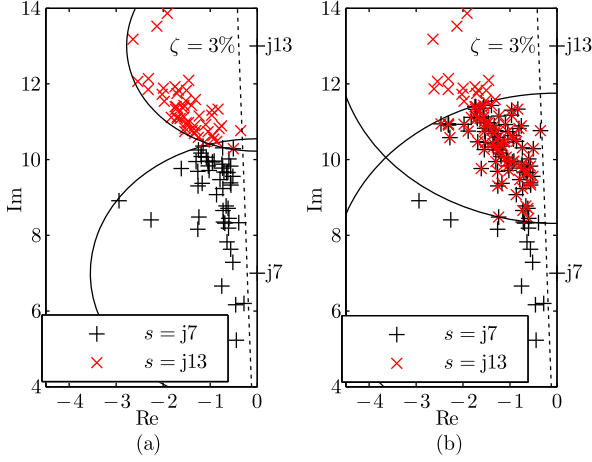


Fig. 8. Clusters of r eigenvalues of $\mathcal{A}_N$ around two shift points $s = j7$ and $j13$ computed by EIGD. (a) $r = 50$. (b) $r = 100$. In (a), the two black arcs are parts of two circles centering about the two shift points with radii of 3.5501 and 2.7781, respectively. Each circle contains 50 eigenvalues which are shown in black + and red x, respectively. In (b), the two black arcs are parts of two circles centering about the two shift points with radii of 4.7569 and 4.6845, respectively. In each circle, 100 eigenvalues are shown in black + and red x, respectively. In (a) and (b), each dotted line is composed of points with constant damping ratio $\zeta = 3\%$.

TABLE III
COMPARISON OF THE THREE METHODS
[# OF ARNOLDI ITERATIONS/CPU TIME (S)]

Methods	s	r	
		50	100
EIGD	j7	7 / 133.87	2 / 160.11
	j13	4 / 103.52	2 / 150.42
IIGD	j7	Failed	Failed
	j13	Failed	Failed
QR		Failed	

In Fig. 8(a), EIGD finds two clusters of eigenvalues centering about the two shift points with radii of 3.5501 and 2.7781, respectively. The two clusters are barely overlapped with only one eigenvalue in common. In Fig. 8(b), EIGD finds two clusters of eigenvalues centering about the two shifts with radii of 4.7569 and 4.6845, respectively. In this case, the two clusters are overlapped with 83 eigenvalues in common. The total number of distinct eigenvalues in Fig. 8(b) is 117, which means all 113 electromechanical oscillation modes in the system are captured with $r = 100$.

All eigenvalues in both Fig. 8(a) and (b) are highly accurate. In fact, when we use them as initial values for Newton correction, it is found that mismatches $\mathbf{f}$ in (15) are all less than the tolerance 10^{-6} . Thus, the number of required Newton iteration is zero. This shows EIGD maintains high accuracy for large transmission system with multiple time delays.

On the contrary, neither QR nor IIGD is capable of handling the system in this case, as seen in Table III. Obviously, QR is not able to deal with $\mathcal{A}_N$ of a high dimension of 29328. IIGD fails because of the prohibitively high requirement of memory.

C. Impact of Uncertainties in Delays

As indicated in [6], time delay are often non-deterministic and can be modeled by random variables in real systems. In this case, the impact of stochastic time delays on the stability

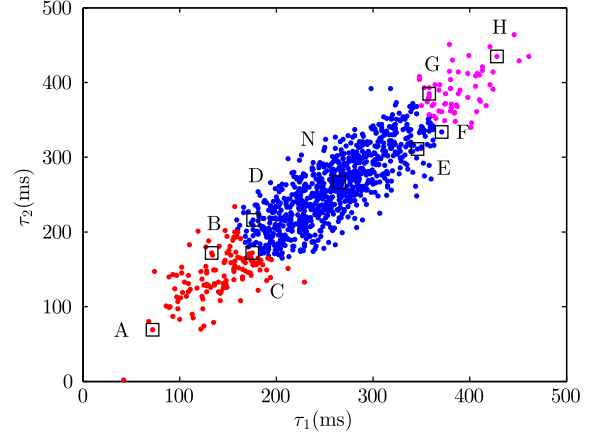


Fig. 9. Random time delays τ_1 and τ_2 . They are denoted as points in different colors. In the lower-left (red) and upper-right (magenta) zones, DCPPS is stable. While in the central zone (blue), DCPPS is unstable. Points A-H and N are randomly selected for illustration.

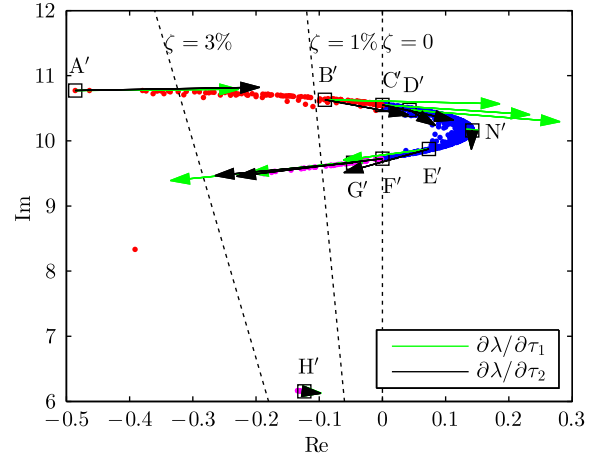


Fig. 10. Rightmost eigenvalues for system under random delays. Eigenvalues in different colors correspond to time delays (points) in the same color in Fig. 9. Especially, eigenvalues A' – H' and N' correspond to A-H and N in the figure, respectively. Green arrows denote eigenvalue sensitivities to τ_1 and black for τ_2 . Three dotted lines are composed of points with constant damping ratios $\zeta = 0, 1\%$ and 3% , respectively.

of the second test system is investigated by conducting 1000 Monte Carlo runs. Here τ_1 and τ_2 in the two channels of the practical large power system are modeled as two identical dependent random variables with mean of 250 ms, covariance of 70 ms [8], and correlation coefficient of 0.9, as shown in Fig. 9. In each Monte Carlo run, only the eigenvalue with the smallest damping ratio ζ is recorded, as shown in Fig. 10. This is because the small signal stability of the DCPPS is determined by this critical eigenvalue.

The whole delay space shown in Fig. 9 can be divided into three zones, corresponding to three segments of the trajectory of critical eigenvalues shown in Fig. 10. In the lower-left zone, the stability of the system deteriorates when τ_1 and τ_2 increase. In the central zone, system loses its stability due to negative damping ratio. However, when the two delays increase and exceed the boundary of the central zone, the system regains stability. This is a surprise finding. The rationale behind it is that the exponential terms in the characteristic equation of DCPPS

can be represented by using trigonometric functions and are inherently periodic.

For illustration purpose, when time delays increase following $A \rightarrow B \rightarrow C(\tau_1 = 175 \text{ ms}, \tau_2 = 172 \text{ ms}) \rightarrow D \rightarrow N(\tau_1 = 265 \text{ ms}, \tau_2 = 267 \text{ ms}) \rightarrow E \rightarrow F(\tau_1 = 371 \text{ ms}, \tau_2 = 334 \text{ ms}) \rightarrow G \rightarrow H$, the trajectory of critical eigenvalues will follow $A' \rightarrow B' \rightarrow C'(j10.5514) \rightarrow D' \rightarrow N'(0.1420 + j10.1571) \rightarrow E' \rightarrow F'(j9.7254) \rightarrow G' \rightarrow H'$.

D. Eigenvalue Sensitivity Analysis

Sensitivities of eigenvalues $A' - H'$ to τ_1 and τ_2 (green arrows for sensitivities to τ_1 and black for τ_2) are computed and shown in Fig. 10. For better illustration, their magnitudes are normalized by 10. It can be seen from the figure that in general, eigenvalue sensitivities coincide with the change of the trajectory of critical eigenvalues. In particular:

- For the upper branch of the trajectory (i.e., $\text{Im} > 10.1571$), eigenvalues $A' - D'$ move towards the right when τ_1 and τ_2 increase and the stability of the system deteriorates. Both magnitudes and angles of eigenvalue sensitivities to τ_1 differ slightly from those to τ_2 .
- At the nose point (eigenvalues N'), it is noted that magnitudes of sensitivities reduce considerably. Besides, the sensitivities point vertically down.
- When τ_1 and τ_2 continue to increase, critical eigenvalues (e.g., $E' - G'$) move towards the left and run to the lower branch of the critical eigenvalue trajectory (i.e., $9 < \text{Im} < 10.1571$). Correspondingly, the stability of the system improves.
- The magnitudes of sensitivities for eigenvalue H' are very small, compared with those of other eigenvalues $A' - G'$. It means that eigenvalue H' is less affected by time delays. When time delays increase, the direction of the movement of H' is right.

VII. CONCLUSION

This paper resolves a long-standing challenging problem of computing critical eigenvalues for large DCPSS by the presented EIGD approach. EIGD features a highly structured, sparse and explicit approximant matrix of infinitesimal generator that is able to reformulate the DDEs of DCPSS into an ODE while maintaining high accuracy in calculating the critical eigenvalues of DCPSS. Furthermore, EIGD exploits the sparsities in both the approximant matrix and system matrices, which allows one to fully utilize sparsity eigenvalue techniques in solving large DCPSS. EIGD therefore guarantees efficiency and scalability for large DCPSS.

EIGD is a highly desirable method for the eigen-analysis of DCPSS such as bulk power systems equipped with wide-area measurement and control systems, microgrids, and active distribution grids with two-way communication and control equipment. In particular, EIGD is useful in determining time delay margins (the maximum time delay under which DCPSS remains stable) in a numerically efficient manner. It can also be used to eliminate or alleviate time delay effects through robust design of wide-area supplementary damping controllers. These will significantly benefit power system operators and planners in operating and managing power grids.

APPENDIX

SPARSE IMPLEMENTATION OF STEPS 5 AND 9 IN ALGORITHM 1

The sparse implementation of step 5, i.e., $\mathbf{p} = \tilde{\mathbf{A}}'_0 \mathbf{v}_j$, is as follows:

$$\begin{aligned}\mathbf{r} &= -\mathbf{C}_0 * \mathbf{v}_j; [\mathbf{L}_1, \mathbf{U}_1, \mathbf{P}_1, \mathbf{Q}_1] = \text{lu}(\mathbf{D}_0); \\ \mathbf{q} &= \mathbf{Q}_1 * (\mathbf{U}_1 \setminus (\mathbf{L}_1 \setminus (\mathbf{P}_1 * (\mathbf{r})))) ; \\ \mathbf{p} &= (\mathbf{A}_0 - s\mathbf{I}_n) * \mathbf{v}_j + \mathbf{B}_0 * \mathbf{q}.\end{aligned}$$

The sparse implementation of step 9, i.e., $\mathbf{q}_{k+1} = (\mathbf{R}'_0)^{-1} \mathbf{z}$, is as follows:

$$\begin{aligned}[\mathbf{L}_2, \mathbf{U}_2, \mathbf{P}_2, \mathbf{Q}_2] &= \text{lu}(\mathbf{A}'(s)); \\ \mathbf{D}^* &= \mathbf{D}_0 - \mathbf{C}_0 * \mathbf{Q}_2 * (\mathbf{U}_2 \setminus (\mathbf{L}_2 \setminus (\mathbf{P}_2 * \mathbf{B}'(s)))) \\ [\mathbf{L}_3, \mathbf{U}_3, \mathbf{P}_3, \mathbf{Q}_3] &= \text{lu}(\mathbf{D}^*); \\ \mathbf{p} &= -\mathbf{C}_0 * \mathbf{Q}_2 * (\mathbf{U}_2 \setminus (\mathbf{L}_2 \setminus (\mathbf{P}_2 * \mathbf{z}))) ; \\ \mathbf{w} &= \mathbf{Q}_3 * (\mathbf{U}_3 \setminus (\mathbf{L}_3 \setminus (\mathbf{P}_3 * \mathbf{p}))) ; \\ \mathbf{r} &= \mathbf{z} - \mathbf{B}'(s) * \mathbf{w}; \\ \mathbf{q}_{k+1}(1:n) &= \mathbf{Q}_2 * (\mathbf{U}_2 \setminus (\mathbf{L}_2 \setminus (\mathbf{P}_2 * \mathbf{r}))).\end{aligned}$$

ACKNOWLEDGMENT

The authors would like to thank Dr. E. Jarlebring from KU Leuven, Belgium, for the helpful discussion. The authors also would like to thank the anonymous reviewers for the valuable comments.

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Hua Ye (M'13) received the B.Eng. and Ph.D. degrees in electrical engineering from Shandong University, Ji'nan, China, in 2003 and 2009, respectively.

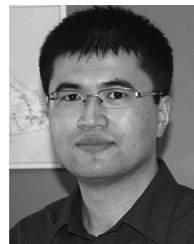
He is an Associate Professor with the Key Laboratory of Power System Intelligent Dispatch and Control of Ministry of Education (Shandong University), Ji'nan, China. His research interests include power system dynamic stability analysis and control. Currently, he is also a Visiting Scholar with the University of Connecticut, Storrs, CT, USA.



Yutian Liu (SM'96) received the B.Eng. and M. S. degrees in electrical engineering from Shandong University of Technology, Ji'nan, China, in 1984 and 1990, respectively, and the Ph.D. degree in electrical engineering from Xi'an Jiaotong University, Xi'an, China, in 1994.

Currently, he is a Professor with the Key Laboratory of Power System Intelligent Dispatch and Control of Ministry of Education (Shandong University), Ji'nan, China. His research interests include power system analysis and control and artificial intelligence

applications in power systems.



Peng Zhang (M'07–SM'10) received the Ph.D. degree in electrical engineering from the University of British Columbia, Vancouver, BC, Canada.

He is an Assistant Professor of Electrical Engineering at the University of Connecticut, Storrs, CT, USA. He was a System Planning Engineer at BC Hydro and Power Authority, Vancouver. His research interests include power system resilience and reliability, microgrid, distributed renewable energy systems, wide-area measurement and control systems and active distribution grid.

Dr. Zhang is a Registered Professional Engineer in British Columbia, Canada.